

Ahmad Sidik Rofiudin

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I'm a Software Engineer with experience in building and maintaining web-based systems, including real-time applications and API integrations. I work across both frontend and backend, and I'm comfortable handling data, system logic, and performance considerations. I enjoy building reliable systems and continuously improving how applications run in production.

WORKING EXPERIENCE

Telkom University – Faculty of Economics and Business, Bandung

Sep 2025 - Present

Software Engineer

- Designed and developed **MIRA (Media Informasi dan Relasi Anda)**, an embedded web platform and AI-powered WhatsApp bot to automate academic administration and operational workflows.
- Devised a **scalable scheduling system** with an interactive drag-and-drop calendar and multi-day event support for enterprise-level academic coordination.
- Executed **real-time conflict detection algorithms** to prevent scheduling overlaps across rooms and faculty members, improving operational reliability.
- Developed a dynamic contract management engine to automate quarterly KPI calculations, reducing manual processing time by **approximately 30–40%** and improving reporting accuracy.
- Integrated an **AI-based WhatsApp bot** to automate administrative tasks such as scheduled notifications and personalized reminders.
- Refactored backend architecture into modular services, improving maintainability and enabling faster feature iteration.

M&A Associates Orthodontics (Orthonova), Tokyo, Japan

Feb 2025 - Sep 2025

AI Engineer

- Architected an AI-powered conversational assistant using Google Gemini to support patient communication and clinical operations in an orthodontic healthcare environment.
- Implemented **context-aware AI workflows** to enable staff in handling patient inquiries with higher accuracy through conversation-driven data retrieval.
- Incorporated doctor-to-doctor communication context into AI pipelines to assist clinical decision support and response generation.
- Built a **natural language search system** leveraging **LangChain SQL Agent**, reducing complex patient data retrieval time by 40%.
- Engineered a **Retrieval-Augmented Generation (RAG)** pipeline to summarize clinical conversations, improving documentation efficiency and reducing manual input by over 50%.
- Unified structured (database) and unstructured (conversation) patient data through context-aware system design, enabling more accurate and consistent AI responses.
- Optimized prompt engineering and system logic to enhance **response accuracy, safety, and multilingual capability** in a healthcare setting.
- Architected scalable frontend architecture using Next.js, Tailwind CSS, and shadcn/ui to deliver a modern and intuitive user interface.

Icommits IT Consultant, Bandung

Jun 2023 – Okt 2023

Fullstack Developer

- Created and developed a **custom Content Management System (CMS)** using **Laravel** to enable dynamic website content management without direct code modification.
- Developed and deployed a **web hosting landing platform**, including multiple customizable pages powered by the CMS for flexible content creation.
- Enabled **dynamic page generation features**, allowing users to create and manage pages with custom headers, text, and media through an intuitive interface.

- Collaborated with the internal IT team to troubleshoot and resolve system issues, ensuring stable and reliable website functionality.

PROJECT EXPERIENCE

Flowbeat – IoT Health Monitoring System

- Engineered a **real-time health monitoring system** integrating IoT devices with a web and mobile dashboard to support early warning detection for cardiac patients.
- Delivered backend APIs using **Laravel** and **JWT authentication** to ensure secure and reliable data transmission between patient devices and clinical dashboards.
- Introduced real-time data processing pipelines to handle continuous sensor streams and trigger alert mechanisms for abnormal conditions.

Resumind – AI-Powered Resume Builder SaaS

- Established a full-stack SaaS application for real-time resume creation with customizable templates and dynamic content editing.
- Orchestrated **AI-assisted content generation** to help users compose work experience and education entries more efficiently.
- Built backend services using **Prisma ORM** and **PostgreSQL**, with user management handled via **Clerk** and subscription system powered by **Stripe**.
- Formulated a tiered subscription model, enabling advanced features such as multiple resumes, styling options, and AI assistance.

Converto – Client-Side File Processing Web App

- Engineered a **browser-based file processing tool** supporting image compression, background removal, and format conversion without server-side storage.
- Achieved up to 70% **reduction in image file size** while maintaining visual quality through optimized client-side processing techniques.
- Crafted a privacy-focused architecture by processing all files locally on the client without database dependency.

EDUCATION

Telkom University , Bandung <i>Master of Information Systems</i> GPA 3.71 / 4.00	Sep 2024 – Feb 2026
Telkom University , Bandung <i>Bachelor of Information Systems</i> GPA 3.83 / 4.00	Aug 2020 – Aug 2024

CERTIFICATIONS

BNSP <i>Certified Web Developer</i>	Dec 2023
HackerRank <i>SQL Advance Certification</i>	Dec 2024

LANGUAGES AND SKILLS

- Bahasa Indonesia (Native), English (Professional Working Proficiency)
- Backend: Node.js, Express.js, Laravel, Python, REST API Design, Prisma ORM, JWT Authentication
- Database: PostgreSQL, MySQL, MongoDB, Firebase, Query Optimization
- AI: LLM Integration, Prompt Engineering, Retrieval-Augmented Generation (RAG), LangChain
- Frontend: Next.js, React, React Native, Typescript, TailwindCSS
- Tools: Git, Postman, Stripe, Clerk, Google OAuth, Docker, GCP, Vercel, Firebase Cloud